

Cal Verdict Application — Relabel List

Grace

2026-05-18

Cal Verdict Application Sweep — Relabel List

Scope

Per Keeper’s autonomous-window assignment 2026-05-18: “Grace: Cal verdict application sweep — relabel list across 105+ papers for T1990/T1992/T2001/T2010/T2071/T2073/22-anomaly.”

Six Cal verdicts from Sunday-pending Task #18-#24 grading: - #18 T1990 $\alpha^2 \cdot 42 \rightarrow$ STRUCTURAL stands (D-tier T1990; recurrence catalog S-tier Type C) - #19 T1992 $r_p \rightarrow$ PROVISIONAL D-tier with caveat (acceptable) - #20 T2001 α^{-1} correction $\rightarrow$ HONEST I-tier (1.4% on correction, not 0.0004% on full) - #22 T2010 mesons + $m_t/m_b=42 \rightarrow$ I-tier mesons / S-tier m_t/m_b ratio (Type C catalog caveat) - #23 T2071+T2073 muon g-2 $\rightarrow$ STRONG I-tier NOT D-tier (0.019% multi-loop convergence; QED-from-D_IV⁵ open) - #24 22-anomaly enumeration $\rightarrow$ REQUIRED per-anomaly tier table before external use

Cal’s load-bearing warning: “If Paper #106 or #108 says ‘BST closes muon g-2 at 0.005%’ without the I-tier qualifier, that ships as crank-class overreach. With the qualifier, it ships as a striking observation inviting investigation.”

Catalog relabels — EXECUTED 2026-05-18

Cal #23 muon g-2 D $\rightarrow$ I relabels (4 entries)

data/bst_geometric_invariants.json:

Index	Entry name	Old tier	New tier	Notes
1641	Muon g-2 HVP closed form	D	I	Cal verdict #23 note appended
2024	HVP closed form verified to 1000 digits (Elie Toy 1663)	D	I	Cal verdict #23 note appended
2105	Muon g-2: a_μ^{HVP} fully D-tier	D	I	Self-declared “fully D-tier” name — Cal anti-crank warning case
2248	Full a_μ from D_IV ⁵ : QED+HVP+HLbL+EW = $116591980 \times 10^{-11}$	D	I	Cal’s specific anti-crank warning case

Standard appended note: “Cal verdict #23 applied 2026-05-18 Grace: STRONG I-tier, NOT D-tier. 0.019% multi-loop convergence is striking observation; QED-from-D_IV⁵ first-principles derivation remains open.”

INV-4401 (B5 Muon g-2 LAG-1 mechanism support) left at D-tier — phrasing explicitly says “mechanism support,” not “value derivation”; defensible at D-tier.

Cal #20 α^{-1} correction honest precision (1 entry)

data/bst_constants.json:

ID	Entry	Old precision	New precision	Action
const_137	Fine structure constant inverse (BST refined)	0.0004%	1.4% (correction term)	Precision label honestified; full value precision was on integer $N_{\text{max}}=137$ baseline, correction-term precision is the falsifiable BST claim

Cal #19 r_p provisional caveat (1 entry)

data/bst_constants.json:

Entry	Action
Proton charge radius	D-tier preserved with PROVISIONAL caveat: “ $\text{rank}^2 \cdot \hbar c / m_p$ form is structurally clean, but PRad-II/MUSE/JUDE convergence remains the empirical falsification gate”

Cal #18 T_{1990} (m_b derivation) clarification (1 entry)

data/bst_constants.json:

Entry	Action
Bottom quark mass (from cascade)	D-tier preserved with Cal #18 qualifier: “BST primary form $C_{2\cdot g}$ (D-tier mechanism) drives this derivation; standalone ‘ $m_t/m_b=42$ recurrence’ framing without $C_{2\cdot g}$ specification stays S-tier Type C catalog”

Cal #22 T_{2010} meson note (4 entries)

data/bst_geometric_invariants.json:

All four T_{2010} meson entries (Kaon-to-pion ratio, B-meson-to-pion ratio, D_s decay constant, INV-4266 $f_K/f_\pi = C_{2/n_C}$) already at I-tier — Cal qualifier appended noting $m_t/m_b=42$ standalone form would be S-tier.

Tier distribution change

Tier	Before	After	Net
D	3387	3383	−4 (muon g-2 $D \rightarrow I$)
I	187	192	+5 (4 muon g-2 + 1 promotion earlier)
S	850	850	0
Catalog total	4426	4427	+1 (Elie B7 closure INV-4427)

Paper-text relabels — NEEDS TEAM COORDINATION

Six claim sites identified across the paper corpus that need Cal I-tier qualifier addition. These are paper-text edits with higher stakes than catalog notes (changes ship to external readers); deferring to Casey/Keeper coordination for the actual edits.

Paper #106 v0.4 — assembled

Line 726-727 (Section 5.5 muon g-2):

“The recurrence extends to a third observable: muon g-2. [...] $\Delta a_\mu \approx \alpha^2 \cdot 42 \cdot$ (small geometric prefactor) $\approx 2.4 \times 10^{-9}$, matching the FNAL world average at sub-percent precision.”

Cal qualifier needed: Append sentence or footnote: *“Tier I per Cal verdict #23: striking 0.019% multi-loop convergence; QED-from-D_{IV}⁵ first-principles derivation remains open. The match is a falsifiable BST claim worth investigating, not a closed structural derivation.”*

Lines 734, 747 (m_t/m_b = 42 recurrence rows):

“m_{top} / m_{bottom} $\approx$ 42” (line 734) “| m_{top}/m_{bottom} | Yukawa heavy-quark mass ratio | 42 | (PDG) | $\sim$ 2% |” (line 747)

Cal qualifier needed (per #22): Footnote on the m_{top}/m_{bottom} row: *“S-tier Type C without C₂-g BST primary form; D-tier only when derivation chain explicitly uses 42 = C₂-g rather than the bare recurrence.”*

Paper #106 section 5 branching ratios draft

Lines 175, 188 — same m_t/m_b=42 claims as v0.4 Paper #106. Same qualifier needed.

Paper #115 v0.5_PRE outline

Line 128: “42 = C₂-g appears in 16+ BST observables: ε_K kaon CP violation, BR(H $\rightarrow\gamma\gamma$) Higgs diphoton, Δa_μ muon g-2 leading α^2 coefficient...”

Cal qualifier needed: When the muon g-2 entry appears in this list, add tier label: *“(Tier I per Cal verdict #23; multi-loop convergence striking but QED-from-D_{IV}⁵ derivation open).”*

Paper #111 Falsification Suite outline v0.1

Line 18: “BST’s predictive content is extensive: muon g-2 closed-form at 0.005%, Higgs branching ratios at <2%, cosmological parameters...”

Cal qualifier needed (HIGH PRIORITY — exactly Cal’s anti-crank warning case): Replace “muon g-2 closed-form at 0.005%” with: *“muon g-2 at 0.005% (Tier I per Cal verdict #23 — striking multi-loop convergence; QED-from-D_{IV}⁵ derivation open)”*.

This is the most important single edit: Paper #111 is a falsification suite explicitly aimed at external readers, and the 0.005% claim without I-tier qualifier is exactly what Cal flagged as crank-class overreach.

Cal verdict #24 — 22-anomaly enumeration (NOT in this sweep)

Cal required per-anomaly tier table before external use. This is a separate enumeration task requiring: - List of 22 anomalies - BST-resolved tier for each (D/I/S/C) - “Resolved” definition per anomaly (closed-form match vs structural identification vs falsifiable correction)

Recommendation: assign to a dedicated session, not folded into this relabel sweep. Affects which observational claims can ship externally.

Affected papers requiring future review (no claim sites flagged in this scan)

10 papers referenced muon g-2 / α^{-1} / r_p but no obvious overclaims found. Listed for completeness:

- Paper #106 v0.3 assembled (predecessor; check if has same claim sites as v0.4)
- Paper #115 v0.1, v0.2, v0.3, v0.3.1, v0.4 (predecessor outlines)
- Paper #118 v0.2 Bergman Dirac
- Paper83 Draft, Outline, Priority_Entries

Summary statistics

- Catalog relabels executed: 11 entries (4 muon g-2 D→I + 1 α^{-1} honest + 1 r_p caveat + 1 m_b clarify + 4 T2010 notes)
- Paper claim sites identified: 6 across 4 papers (#106 v0.4, #106 sec 5, #115 v0.5_PRE, #111)
- High-priority single edit: Paper #111 line 18 (Cal's specific anti-crank warning case)
- Deferred: Cal #24 22-anomaly enumeration (separate dedicated session)

Authority and coordination

Catalog relabels are within Grace's data/ lane authority — applied directly.

Paper-text relabels are higher-stakes (external-facing) and require Keeper coordination + Casey approval before shipping. List filed; awaiting Casey direction on whether to proceed with paper edits in this session or defer.

— Grace, Cal Verdict Application Sweep, 2026-05-18 ~16:00 EDT